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Values Are Not Enough: Situational Modifiers Shape Value-Based Decisions in LLMs and Humans

Key Highlights

- Research Problem: Values alone do not predict moral choices; situational modifiers can shift decisions in humans and LLMs.
- Proposed Solution: Create VISDA (80 scenarios, 8 modifier axes) and a decision-flip metric to measure context-induced choice changes across value profiles.
- Unique Contribution: Evaluate 3 LLMs, a utility baseline (95 Schwartz profiles), and humans; reveal LLM miscalibration favoring personal-cost modifiers.

Literature Review

Literature Review Analysis

Search Keywords for Cited Papers

- Value-driven moral decision-making
- Situational modifiers and person-situation dynamics
- Schwartz value theory and PVQ-21
- LLM moral reasoning benchmarks
- Open-weight LLM evaluation (Gemma, Qwen, LLaMA)

Most Important Cited Papers

- Gemma Team, Gemma: Open models based on gemini research and technology, 2024: Introduces an open model family relevant for evaluating model behavior in value- and context-sensitive settings.
- Bai et al., Qwen technical report, 2023: Describes the Qwen model family, a commonly referenced open-weight foundation for empirical evaluation.
- Touvron et al., LLaMA: Open and efficient foundation language models, 2023: Presents the LLaMA family widely used in downstream moral reasoning and alignment studies.
- Schwartz et al., Refining the theory of basic individual values, 2012: Provides the refined value theory that underpins the reduced value dimensions employed for constructing value profiles.
- Shalom H Schwartz, A proposal for measuring value orientations across nations. Questionnaire Package of the European Social Survey, 2003: Introduces the PVQ-21 instrument used for measuring Schwartz values.
- Shalom H Schwartz, Universals in the content and structure of values: Theoretical advances and empirical tests in 20 countries, 1992: Establishes the foundational Schwartz value framework used to define basic human values.

Timeliness of Cited Papers

- Most recent studies after 2024: Multiple works from 2025-2026 are cited, including Guo et al. (2026), Harry et al. (2026), Nabizadeh et al. (2026), Sauter and Schirmer (2026), and Backmann et al. (2025).
- Recent studies (2022-2023): Key sources include Abdulhai et al. (2023), Bai et al. (2023), Durmus et al. (2023), Kang et al. (2023), and Santurkar et al. (2023).

- The literature review includes recent developments after 2022, spanning both methodological and evaluative fronts: it cites Sorensen et al. (2024) on pluralistic values, Santurkar et al. (2023) and Durmus et al. (2023) on opinion/priming-induced shifts, Sauter and Schirmer (2026), Backmann et al. (2025), and Nabizadeh et al. (2026) on contextual effects in moral judgments, and Guo et al. (2026) and Jiao et al. (2025) on steering via values/personality. This shows strong coverage of current LLM moral reasoning and context-sensitivity research.
- The sources reflect the current state of knowledge by integrating up-to-date model families and benchmarks alongside contemporary alignment debates: inclusion of Gemma Team (2024) and Bai et al. (2023) demonstrates evaluation on recent open-weight LLMs; Touvron et al. (2023) anchors the LLaMA family; Harry et al. (2026) highlights state-over-trait dynamics in LLMs; and Samway et al. (2025), Mahajan (2025), and Schacht and Lanquillon (2025) situate VISDA relative to large-scale moral reasoning analyses. Together, these citations align with the evolving understanding that situational context significantly shapes LLM moral outputs beyond fixed trait/value prompts.

Relevance of Cited Papers

- Direct link to value frameworks guiding behavior: The manuscript grounds its framework in Schwartz's value theory, using Schwartz (1992) and Schwartz et al. (2012) as the theoretical basis for representing values and PVQ-21 (Schwartz, 2003) as the measurement instrument. This directly supports their design of matched value profiles and the utility baseline over Schwartz profiles.
- Direct examination of value-conditioned behavior in LLMs: Prior studies such as Sorensen et al. (2024), Abdulhai et al. (2023), Scherrer et al. (2023), Kang et al. (2023), Guo et al. (2026), and Jiao et al. (2025) probe or steer LLMs via values and personalities. These works establish that altering value profiles or moral beliefs yields predictable behavioral shifts, aligning with the manuscript's premise that values can steer actions and motivating a test of situational sensitivity under fixed profiles.
- Direct evidence for situational/contextual effects: Santurkar et al. (2023) and Durmus et al. (2023) show outputs shift under demographic/geographic priming; Sauter and Schirmer (2026), Backmann et al. (2025), and Nabizadeh et al. (2026) demonstrate contextual framing, opponent behavior, and survival risk alter moral judgments in LLMs. These substantiate the core research question that situational modifiers influence decisions beyond fixed values.
- Comprehensive background on the person-situation debate: Classic social psychology foundations (Mischel, 1968; Ross and Nisbett, 1991) and recent evidence of within-person variance (Harry et al., 2026) provide a theoretical backdrop that behavior depends jointly on dispositions and context, framing why fixed value profiles may not fully predict actions. This positions VISDA within established psychological debates adapted to LLM alignment.
- Context within moral reasoning benchmarks and gaps: ETHICS (Hendrycks et al., 2021), MoralStories (Emelin et al., 2021), Samway et al. (2025), Mahajan (2025), and Schacht and Lanquillon (2025) are cited as large-scale moral reasoning resources. The manuscript clearly articulates their limitation for the present question - they do not systematically vary situational modifiers under fixed value profiles - thereby motivating the need for VISDA.

Suggested Improvements

Key Action Points

- Clarify the theoretical rationale for choosing Schwartz values over moral foundations: Expand the comparison between Schwartz's circumplex and Haidt's moral foundations, specifying how action-level mapping and measurement via PVQ-21 justify design decisions, while acknowledging complementary coverage of moral domains. Include explicit citations to Haidt (2012) and the PVQ-21 description.
- Integrate recent person-situation evidence in LLMs to strengthen the framing: Situate the study's within-profile, across-context design alongside findings that LLM outputs are highly state-sensitive and can be altered by situational context, drawing on Harry et al. (2026) and Peterson (2025) to connect classic person-situation theory to alignment evaluation.
- More comprehensively synthesize the value-steering literature: Summarize convergent findings across Sorensen et al. (2024), Kang et al. (2023), Guo et al. (2026), Abdulhai et al. (2023), Scherrer et al. (2023), and Jiao et al. (2025), explicitly distinguishing prompt-based value injection, counterfactual steering, and inferred moral belief analyses, and explain how VISDA isolates situational effects under fixed profiles relative to these paradigms.
- Deepen coverage of contextual priming and framing effects: Connect demographic and geographic priming results to moral decision shifts by summarizing Santurkar et al. (2023) and Durmus et al. (2023) alongside recent moral dilemma variations from Sauter and Schirmer (2026), Backmann et al. (2025), and Nabizadeh et al. (2026), and specify which situational modifiers VISDA operationalizes relative to these prior manipulations.
- Clarify how the benchmark differs from existing moral reasoning datasets: Explicitly contrast the within-profile manipulation in VISDA with ETHICS and MoralStories, and with analytical frameworks in Samway et al. (2025),

Mahajan (2025), and Schacht and Lanquillon (2025), highlighting the absence of systematic situational variation under fixed value profiles in prior work.

- Strengthen relevance by explicitly tying model family selection to research aims: When referencing LLaMA, Qwen, and Gemma, add brief justification linking model characteristics to the study's need to probe value-conditioned and context-sensitive outputs, with precise citations to the technical reports.
- Bridge classic social psychology with current LLM findings: Explicitly connect Mischel (1968) and Ross and Nisbett (1991) to the LLM-specific evidence on state sensitivity and context dependence, situating VISDA as a controlled within-subject design that operationalizes person-situation dynamics for models.

Recommended Papers to Read and Cite

Recent Publications to Improve Timeliness

- Yinpei Su et al., Understanding How Value Neurons Shape the Generation of Specified Values in LLMs, 2025. [Link](#)
- Andrey Elster et al., Personal Values and Cognitive Biases, 2024. [Link](#)
- Oscar Smallenbroek et al., Constructing Schwartz values framework using the Rokeach values survey: Human value measurement in the longitudinal internet survey for social sciences, 2025. [Link](#)
- Nadav Borenstein et al., Investigating Human Values in Online Communities, 2024. [Link](#)
- Sanqing Qu et al., LEAD: Learning Decomposition for Source-free Universal Domain Adaptation, 2024. [Link](#)
- Lauren E. Mullenbach et al., Values outweigh political ideology when forming beliefs about gentrification: A study of the US general public, 2025. [Link](#)
- Zhiqi Yu et al., Dynamic Target Distribution Estimation for Source-Free Open-Set Domain Adaptation, 2025. [Link](#)
- A. Sandibekova et al., Pedagogical management to improve environmental education in the context of sustainable development, 2024. [Link](#)
- Mingxuan Xia et al., A Separation and Alignment Framework for Black-Box Domain Adaptation, 2024. [Link](#)
- Andrea Vega-Tinoco et al., Human values, civic participation, and wellbeing: analysis on their relationship among older Europeans, 2024. [Link](#)

Relevant Papers to Address Research Gaps

- Loes Abrahams et al., Person-situation dynamics in educational contexts: A self- and other-rated experience sampling study of teachers' states, traits, and situations, 2021. [Link](#)
- D. Hadar-Shoval et al., Embedded values-like shape ethical reasoning of large language models on primary care ethical dilemmas, 2024. [Link](#)
- Mohna Chakraborty et al., Structured Moral Reasoning in Language Models: A Value-Grounded Evaluation Framework, 2025. [Link](#)
- José Luiz Nunes et al., Are Large Language Models Moral Hypocrites? A Study Based on Moral Foundations, 2024. [Link](#)
- Yu Ying Chiu et al., DailyDilemmas: Revealing Value Preferences of LLMs with Quandaries of Daily Life, 2024. [Link](#)
- Jacques P. de Wet et al., Revisiting the trans-situationality of values in Schwartz's Portrait Values Questionnaire, 2018. [Link](#)
- Ya Wu et al., The Staircase of Ethics: Probing LLM Value Priorities through Multi-Step Induction to Complex Moral Dilemmas, 2025. [Link](#)
- Heena Rathore et al., Assessing Gender and Age Influences on Moral Decision Making in Autonomous Vehicles Using Large Language Models, 2025. [Link](#)
- Bumjin Park et al., Deontological Keyword Bias: The Impact of Modal Expressions on Normative Judgments of Language Models, 2025. [Link](#)
- Younghun Lee et al., Towards Characterizing Subjectivity of Individuals through Modeling Value Conflicts and Trade-offs, 2025. [Link](#)

Methodology & Solution

Methodology Analysis

Overall Study Design

- Research objective and hypothesis are clearly articulated: assess whether situational modifiers change decisions when Schwartz value profiles are held fixed, via VISDA and a decision-flip metric. This design directly operationalizes the claim that values alone do not determine actions.
- Comparative design includes a utility-based mathematical baseline, three open-weight LLMs (LLaMA 3.1 8B, Gemma 4 31B, Qwen 2.5 32B), and a human pilot, all evaluated under the same scenario framework and fixed value profiles, enabling cross-system triangulation of situational sensitivity.
- Within-subject pairing for LLMs permits clean flip measurement per (value profile, scenario, modifier) tuple; however, the human pilot uses a between-subject modifier exposure, limiting direct comparability of flip metrics across humans and LLMs.
- Counting inconsistency remains: the abstract and section 3.2/3.3 reference a dataset of 80 binary moral scenarios, yet section 3.2(ii) defines 10 scenarios (two per theme) with 8 modifiers each, and section 3.3 states "80 scenario-modifier pairs." The LLM baseline count further reports $3 \times 10 \times 95 = 2,850$ baseline decisions but later pairs $80 \times 95 = 7,600$ modified trials per model, creating a mismatch in baseline pairing logic. Clarification is required to ensure reproducibility.

Data Collection Process

- Dataset construction specifies 5 themes and 10 scenarios (two per theme), each with two actions (A0/A1) and 8 modifiers across enumerated axes; modifiers are annotated with expected_value_pressure for the utility baseline. The manuscript also states "80 binary moral scenarios," which conflicts with the 10-scenario definition and appears to actually denote 80 scenario-modifier pairs; this needs explicit correction.
- Value profiles: 95 curated binary profiles over 10 Schwartz values are selected by pruning antagonism-violating combinations and sampling for coverage and Hamming diversity; natural-language profile descriptions are prepared for prompting. Full profile list and curation details are deferred to an appendix.
- LLM decision collection: each model is evaluated on 80 scenario-modifier pairs across 95 profiles ($3 \times 80 \times 95 = 22,800$ modified decisions total), plus baseline scenarios across profiles. The reported baseline total ($3 \times 10 \times 95 = 2,850$) is numerically inconsistent with subsequent pairing of 7,600 modified trials per model, obscuring exact baseline-modifier pairing counts.
- Human pilot: 50 participants completed PVQ-21, six forced-choice value trade-offs, and SDS-10; approximately 500 decisions were recorded. Each scenario was seen once by a participant either in baseline or in the strongest-LLM-flip modifier condition for that scenario. The manuscript text also includes an inconsistent sentence implying "decisions for both (A1) and (A0) settings were recorded," which conflicts with the between-subject design and needs clarification. Recruitment, demographics, compensation, and IRB details are not reported.

Devices, Software, & Tools

- LLMs and compute: LLaMA 3.1 8B-Instruct, Gemma 4 31B-Instruct, Qwen 2.5 32B-Instruct evaluated locally on a cluster of 4 NVIDIA A100 GPU nodes with greedy decoding (temperature = 0) to ensure deterministic outputs. This setup is suitable for reproducibility.
- Utility-based mathematical baseline: defined with manual action vectors VA_i , modifier perturbations δm derived from expected_value_pressure, and $\lambda = 0.5$ to model moderate contextual influence. Equations for utility and flip definition are provided.
- Prompting pipeline: prompts include the value-profile description, scenario text, modifier_text for modified trials, and actions A0/A1; the model must respond with a single token A0 or A1, minimizing parsing ambiguity. Exact prompt templates are referenced but not included in the provided text.

Experimental Procedure

- Scenario construction applies four constraints: plausibility and social acceptability, value-priority contrasts rather than outcome quality, normalized wording across siblings, and checks against lexical leakage; items with overly one-sided choices or transparent modifier effects were removed after a pilot pass.
- Modifier annotation: each modifier is annotated with expected_value_pressure and disagreements are resolved by consensus; annotations inform only the utility baseline, while LLMs and humans see only natural-language modifiers.
- LLM evaluation: paired baseline vs. modifier outcomes for the same model, scenario, and value profile are used to compute flips; decoding is deterministic. However, the reported baseline count vs. pairings is inconsistent, and prompt templates are missing, limiting exact replication.
- Human pilot procedure: between-subject exposure to modifiers (baseline or strongest-LLM-flip modifier per scenario). This design precludes within-subject flip measurement and leads to pooled shift analyses; exact task

timing, randomization, and counterbalancing details are not reported.

Data Analysis Methods

- Primary metric: decision-flip indicator $F_s(\text{VSW}, S, M) = I[\text{fs}(\text{VSW}, S, \text{I}) \neq \text{fs}(\text{VSW}, S, M)]$ averaged over tuples, directly capturing situational sensitivity under fixed values.
- Statistical testing: McNemar's exact test per (model, axis) with Benjamini-Hochberg FDR correction across eight axes tests directional flips; additional statistical details are in an appendix not provided here.
- Cross-model rank comparison: Spearman correlations over axis orderings show moderate agreement between $\geq 30\text{B}$ models ($\rho = 0.66$, $p = 0.076$), with lower correlations for LLaMA 8B and weak/negative correlations for the utility baseline relative to LLMs, supporting convergent but non-additive behavior.
- Robustness to paraphrase: four meaning-preserving paraphrases for 95 baseline cells per model quantify a noise floor; numerical rates are not reported in the excerpt, limiting assessment of margin beyond prompt variance.
- Aggregation by modifier type: grouping axes into stakes, affective, personal-cost, and informational categories yields interpretable system-level patterns and human-LLM divergence in sensitivity.

Validity & Reliability of Methods

- Internal validity: deterministic decoding (temperature 0), lexical leakage checks, and paraphrase controls reduce stochasticity and prompt artifacts in LLM evaluations.
- Construct validity: the utility baseline depends on manually specified action vectors VA_i and modifier vectors δ_m built from expected_value_pressure; these reflect researcher priors. The paper acknowledges modifier authorship bias and suggests future crowdsourced or LLM-augmented generation.
- External validity: human participants are a convenience sample from a single educational/cultural context; the study adopts the 10-value Schwartz framework rather than the refined 19-value theory, constraining generalization and granularity of value distinctions.
- Reliability concerns: inconsistency in scenario counts and baseline decision accounting (10 scenarios vs. "80 scenarios" vs. "80 scenario-modifier pairs," and 2,850 baseline LLM decisions vs. 7,600 pairings per model) impairs traceability and reproducibility; missing appendices with full profile lists, action/modifier vectors, and prompt templates further limit replication.
- Human-LLM comparability: between-subject human design yields pooled shifts rather than paired flips, and missing procedural details (randomization, timing) could introduce uncontrolled variance in human data.

Proposed Solution Analysis

Clarity & Relevance

- VISDA is defined with five components: themes, scenarios, modifiers, value profiles, and profile descriptions, clearly structured to manipulate situational context while holding value profiles fixed. Modifier axes are explicitly enumerated and linked to expected value pressures for the utility baseline. This directly targets the research question of situational sensitivity under fixed values.
- The evaluation pipeline instructs models to output a single-token choice (A_0 or A_1), reducing ambiguity, and computes a decision-flip metric for within-subject comparison. This operational clarity increases interpretability of situational effects.

Innovation & Impact

- Novelty stems from varying situational context while holding Schwartz value profiles fixed, contrasting with prior moral benchmarks that vary values or lack systematic modifier manipulation under fixed profiles.
- Impact is supported by quantitative findings: utility baseline flips on 19.93% of trials; LLM flip rates are lower at 2.03%-8.92% with a scale effect; humans show pooled shift $|\Delta P(A_1)| = 12.36\%$. Axis-level results emphasize authority_signal and self_preservation for LLM flips, with a strong cell SC004_2 + self_preservation (42.1% Gemma; 31.6% Qwen), revealing actionable miscalibrations relative to human sensitivity.
- The "impossible flips" result and profile-strength moderation demonstrate departures from an additive value-vector model, indicating non-additive contextual reasoning in LLMs when modifiers act as premises, advancing methodological understanding.

Feasibility of Implementation

- Computational feasibility is demonstrated: open-weight models evaluated locally with deterministic decoding; 22,800 modified LLM decisions and per-model pairings are tractable for modern research infrastructure.

- Dataset scope is manageable: 95 curated value profiles and 8 modifiers per scenario enable systematic coverage without excessive annotation burden; human pilot collected approximately 500 decisions as a demonstration.
- Feasibility risks remain in manual specifications: VAI and δm rely on researcher annotations and consensus, which may scale with added scenarios/modifiers, though constraints and checks are described to control leakage and bias.

Effectiveness in Solving Problem

- The decision-flip metric aligns with the core claim and is complemented by per-axis rates, heatmaps, and modifier-type grouping, exposing distinct sensitivity patterns across systems.
- McNemar's exact tests with BH correction support directional flips at the axis level, reinforcing that effects are not random.
- Specific quantitative exemplars strengthen effectiveness: authority_signal and self_preservation are consistently top axes across models, with cross-model Spearman $\rho = 0.66$ ($p = 0.076$) for $\geq 30B$ models on axis ordering; SC004_2 + self_preservation reaches flip rates of 42.1% (Gemma) and 31.6% (Qwen).
- Paraphrase noise control exists but without reported rates in the excerpt; absence of these numbers weakens claims about the margin between observed flips and the prompt-instability floor.

Scalability & Generalizability of Solutions

- Scalability: the framework naturally extends to additional profiles, scenarios, axes, and models, with moderate computational demands and a clear prompting interface for both open- and potentially closed-weight systems.
- Generalizability limits: only the 10-value Schwartz framework is used, human sampling is a convenience cohort from a single context, and only open-weight models are evaluated; authors explicitly note these as limitations affecting cultural transfer, granularity of values, and coverage of frontier models.

Suggested Improvements

Key Action Points

- Resolve scenario and trial-count inconsistencies with a consolidated accounting: explicitly define that the dataset contains 10 scenarios with 8 modifiers each (80 scenario-modifier pairs) and correct all references where the abstract and methods describe "80 binary moral scenarios." Align baseline counts so that per-model baseline trials match the 80×95 modified pairings (e.g., report $80 \times 95 = 7,600$ baseline trials per model if pairing all modified cells), and provide a table that traces, for each model, the exact numbers of baseline and modified trials and pairings.
- Convert the human pilot to a within-subject flip design directly comparable to LLMs: for each participant, present a baseline and one modifier version of the same scenario, counterbalance order, and compute per-participant flip indicators. Increase per-axis sample sizes beyond approximately 500 decisions to enable reliable McNemar testing and confidence intervals for human flip rates alongside LLMs on the same scenario-axis cells.
- Clarify human-protocol text and report administration details: remove the contradictory phrase "decisions for both (A1) and (A0) settings were recorded" in light of the between-subject design; document randomization, counterbalancing, task timing/duration, attention checks, exclusion criteria, recruitment procedures, demographics, compensation, and IRB/ethics approvals to improve interpretability and transparency.
- Release replication materials referenced as appendices: provide the full list of 95 value profiles, manual action vectors VAI, modifier vectors δm with construction from expected_value_pressure, exact prompt templates used for baseline and modified trials, and the λ sensitivity analysis settings for the utility baseline. Include inter-annotator agreement statistics for modifier annotations and details of consensus resolution.
- Report full paraphrase noise results and augment robustness checks: provide per-model paraphrase-induced flip rates for the four paraphrases over the 95 baseline cells; add multiple paraphrase generators and compute confidence intervals so that the excess of modifier-induced flips over the noise floor is quantitatively established.
- Strengthen statistical reporting with uncertainty quantification: include per-axis and per-scenario effect sizes with confidence intervals, exact post-FDR McNemar p-values, and bootstrap or Bayesian estimates of flip-rate uncertainty for each model and axis, enabling rigorous cross-system comparisons.
- Expand modifier generation and validation to mitigate authorship bias: implement a crowdsourced or LLM-augmented generation pipeline across the eight axes with adversarial validation to detect unintended lexical cues and to diversify pressures; document validation metrics and rejection criteria.
- Extend the value taxonomy or map to refined categories: add a follow-up experiment using Schwartz's 19-value refined theory, or provide a mapping analysis showing that the eight modifier axes exert consistent effects when disaggregated into refined values (e.g., humility, face, distinct conformity facets).

- Deepen model-behavior interpretation on high-impact cells: perform qualitative analysis of LLM rationales on cells like SC004_2 + self_preservation where flips reach 42.1% (Gemma) and 31.6% (Qwen), linking reversals to premise uptake and discourse features to substantiate the non-additive contextual reasoning claim beyond aggregate metrics.
- Conduct sensitivity analyses for the utility baseline: vary λ around 0.5 and explore alternative utility forms (e.g., non-linear interactions, value-weighted norms) to test robustness of the 19.93% flip rate baseline and to benchmark the gap between additive and observed LLM behaviors across profile strengths.
- Broaden model coverage and alignment comparison: include closed-weight frontier models (e.g., GPT-4-class systems) to test whether axis-level miscalibrations and scale trends persist, addressing the manuscript's limitation regarding open-weight-only evaluations.
- Align human-LLM exposure for direct comparability: for each scenario-axis pair, collect human baseline and modifier responses from the same individuals or matched cohorts to compute axis-specific human flip rates, enabling one-to-one comparisons with LLM within-subject flips rather than pooled shifts.

Recommended Papers to Read and Cite

Relevant Papers to Strengthen Methodology

- Chris Shaner et al., Assessing Moral Decision Making in Large Language Models, 2025. [Link](#)
- Vanessa Cheung et al., Large language models show amplified cognitive biases in moral decision-making, 2025. [Link](#)
- Karina Vida et al., Decoding Multilingual Moral Preferences: Unveiling LLM's Biases Through the Moral Machine Experiment, 2024. [Link](#)
- Pratik S. Sachdeva et al., Normative Evaluation of Large Language Models with Everyday Moral Dilemmas, 2025. [Link](#)
- M. Ahmad et al., Large-scale moral machine experiment on large language models, 2024. [Link](#)
- Julia Haas et al., A roadmap for evaluating moral competence in large language models, 2026. [Link](#)
- Mohna Chakraborty et al., Structured Moral Reasoning in Language Models: A Value-Grounded Evaluation Framework, 2025. [Link](#)
- Andrea Polonioli, Moving LLM evaluation forward: lessons from human judgment research, 2025. [Link](#)
- Junfeng Jiao et al., LLM ethics benchmark: a three-dimensional assessment system for evaluating moral reasoning in large language models, 2025. [Link](#)
- Divij Bajaj et al., Evaluating Gender Bias of LLMs in Making Morality Judgements, 2024. [Link](#)

Results & Findings

Results & Findings Analysis

Alignment with Research Problem

- The results operationalize the central hypothesis that situational context alters value-driven moral choices, even when Schwartz value profiles are held constant, by introducing an 80-scenario $\times$ 8-axis modifier benchmark and a decision-flip metric. System-level flip rates show nontrivial shifts across systems: utility baseline 19.93% (1,515/7,600), LLMs 2.03%-8.92% (LLaMA 3.1 8B = 154/7,600; Qwen 2.5 32B = 500/7,600; Gemma 4 31B = 678/7,600), and a human pooled $|\Delta P(A1)| = 12.36\%$ in a between-subjects rotation, demonstrating that modifiers affect decisions beyond stable value profiles.
- Observed outcomes match the expected direction but reveal model-specific weighting differences: LLMs show lower overall flip rates than the additive utility baseline, consistent with non-additive integration, yet concentrate flips on authority_signal and self_preservation, whereas humans emphasize in_out_group, authority_signal, and resource_scarcity. This pattern substantiates the stated objective of detecting axis-level miscalibration in situational sensitivity.
- Profile-strength moderation further aligns with the research question by showing that, where the additive baseline is locked to 0% flips for strength-8/9 value profiles by construction, LLMs still flip (e.g., Gemma 20.3%/15.6%; Qwen 10.3%/5.0%; LLaMA 0.6%/0.0%), confirming context-driven behavior beyond value arithmetic.

Accuracy & Precision

- Flip-rate denominators and numerators are explicitly reported for each LLM (7,600 modified trials = 95 profiles $\times$ 80 pairs), with precise counts and percentages (e.g., Gemma 678/7,600 = 8.92%). The human metric is carefully

adapted to the between-subjects design by reporting pooled $|\Delta P(A1)|$ across 10 scenario-axis cells, with an interpretable mapping to about 31 expected response changes out of 250 modified trials.

- A paraphrase noise-floor control uses four meaning-preserving paraphrases for a stratified sample of 95 baseline (VSW, scenario) cells per model, holding values, actions, response format, and decoding fixed. Noise rates are substantially below modifier-induced flip rates (Gemma 2.95% vs 8.92%; Qwen 2.64% vs 6.58%; LLaMA 0.00% vs 2.03%), with two-proportion z-tests rejecting equality ($z > 4.5$, $p < 1e-5$). Paraphrase flips localize to a single scenario per model, increasing confidence that measured flips reflect situational modifiers rather than prompt variance.
- Per-axis directional effects are evaluated using McNemar's exact test on paired baseline-vs-modifier outcomes with Benjamini-Hochberg FDR correction across eight axes, an appropriate nonparametric test for binary paired data. The text states `authority_signal` and `self_preservation` are significant under this procedure; while exact p-values are placed in the appendix, the test choice is methodologically sound.
- Cross-model axis-ordering consistency is quantified with Spearman rank correlations ($\rho = 0.66$ between Gemma 31B and Qwen 32B, $p = 0.076$), with LLaMA 8B showing lower correlations, and the utility baseline weakly or negatively correlated with LLMs ($\rho \in [-0.13, 0.35]$). Reporting both rank order and p-values strengthens precision in claims of convergence and divergence.

Aspects of Research Problem Addressed

- The analysis spans system-level flip rates, a paraphrase robustness control, per-axis flip rates with inferential testing, cross-model consistency, per-(scenario, axis) heatmaps, profile-strength moderation, a formal argument for non-additive contextual reasoning including impossible flips, modifier-type category aggregation, and a human pilot comparison. This multi-angle coverage addresses situational effects comprehensively.
- Granular per-(scenario, axis) results identify concentrated sensitivities, notably SC004_2 + `self_preservation` as the strongest cell for both Gemma (42.1%) and Qwen (31.6%), highlighting non-uniform distribution of effects across the scenario-pressure space.

Clarity in Presentation

- Clarity is enhanced through a schematic of the decision-flip setup (Figure 1), a comprehensive system-level summary (Table 1), a spider plot of per-axis flip rates overlaid with human data (Figure 2), per-(scenario, axis) heatmaps for two LLMs (Figure 3), and modifier-type aggregation with both a figure and a table (Figure 4 and Table 2). Together these visuals support interpretability at multiple granularities.
- The narrative progresses logically from establishing a noise floor to per-axis analysis with inferential tests, followed by cross-model ordering, heatmaps, profile-strength effects, and categorical aggregation, culminating in the human comparison. The manuscript clearly communicates that `authority_signal` and `self_preservation` dominate LLM flips while human sensitivity peaks on `in_out_group` and `resource_scarcity`, supporting the miscalibration claim.

Implication of Findings

- The results reframe alignment evaluation from value-expression checks to axis-specific weighting of situational pressures: LLMs exhibit suppressed overall flipping relative to an additive utility model yet redistribute weight toward `self_preservation` and `authority_signal` compared to human reference patterns, implying targeted calibration is needed at the axis level.
- Cross-pipeline convergence on `authority_signal` and `self_preservation` as top movers aligns with classic social-psychological factors (obedience, self-interest), suggesting that learned norms may dampen some flipping but introduce vulnerabilities to specific cues relevant for red teaming and adversarial prompt design.

Future Work

- Extend evaluation to closed-weight frontier models (e.g., GPT-4, Claude, Gemini) to assess whether the stakes vs personal-cost miscalibration and axis rankings persist at larger scales and under different alignment regimes.
- Broaden and stratify human sampling beyond a single educational/cultural context to test the generality of axis-sensitivity patterns, given the current convenience sample limitation.
- Adopt the refined 19-value Schwartz taxonomy and continuous value encodings to capture finer-grained value-modifier interactions currently collapsed in the 10-value framework and binary representation.
- Implement crowdsourced or LLM-augmented modifier generation with adversarial validation to mitigate selection bias in modifier authorship and probe generalization to unseen pressure phrasings.
- Revisit pruning rules and antagonism-violation handling while moving beyond binary encodings to reduce artificial constraints on the profile space and better model real-world value trade-offs.

Comparative Analysis with Cited Papers

Grounds for Comparison

- Comparisons span three independently trained open-weight LLMs, an additive utility-based mathematical baseline over 95 Schwartz profiles, and a human pilot using the same scenario–modifier design, providing methodologically aligned baselines for situational sensitivity assessment.
- Internal comparisons are strengthened by statistical measures of axis-ordering agreement (Spearman $\rho = 0.66$ for Gemma 31B vs Qwen 32B; LLaMA 8B lower) and by per-axis inferential testing (McNemar with BH correction), establishing robust within-paper benchmarks despite the lack of external numeric baselines from cited literature in the provided text.
- The human pilot offers an ecologically relevant reference distribution for axis-level sensitivity (e.g., peak on in_out_group 21.2%, authority_signal 18.1%, resource_scarcity 16.3%), enabling a direct, controlled contrast with LLMs and the utility baseline within the same experimental framework.

Improvements & Contradictions

- Improvement: LLMs reduce excessive flipping predicted by the additive utility baseline (LLMs: 2.03%-8.92% vs utility 19.93%), indicating non-additive integration that aligns better with stable decision patterns under small modifier perturbations.
- Improvement: Modifier-induced flips significantly exceed paraphrase noise (e.g., Gemma 8.92% vs 2.95%; Qwen 6.58% vs 2.64%; LLaMA 2.03% vs 0.00%), with strong statistical separation ($z > 4.5$, $p < 1e-5$), validating that measured effects reflect situational pressures rather than prompt instability.
- Improvement: Cross-pipeline convergence on authority_signal and self_preservation as the top two axes across LLMs (identical 1/2 or 2/1 ordering; $\rho = 0.66$ between $\geq 30B$ models) provides a replicated internal benchmark of axis-level sensitivity.
- Contradiction: Relative to the human pilot, mid-size LLMs overweight personal-cost/self_preservation and underweight stakes/resource_scarcity and in_out_group; humans show higher mean shifts for stakes (16.3%) and affective modifiers (15.3%) than LLMs (e.g., Gemma stakes 7.8%, affective 9.9%), indicating directional miscalibration.
- Contradiction: The additive utility baseline's per-axis ordering aligns more closely with the human ordering (Spearman $\rho = +0.60$) than any LLM ($\rho = +0.16$ Gemma, -0.05 Qwen, $+0.29$ LLaMA 8B; all $p > 0.1$), challenging expectations that aligned LLMs should mirror human axis sensitivities.
- Contradiction: For strong value profiles (8–9 active values), the utility baseline flips 0% by construction, yet LLMs still flip (e.g., Gemma 20.3%/15.6%; Qwen 10.3%/5.0%; LLaMA 0.6%/0.0%), revealing non-additive contextual reasoning that contradicts a value-arithmetic account.
- Contradiction: The utility baseline ranks authority_signal and in_out_group highest, whereas LLMs rank authority_signal with self_preservation, diverging from both the baseline and human emphasis on in_out_group, underscoring a systematic redistribution of axis weights in LLMs.

Suggested Improvements

Key Action Points

- Broaden human sampling with a stratified design across age bands, professions, and cultural backgrounds to validate axis-level sensitivity patterns. Target per-axis precision by pre-registering power analyses for the 10 scenario–axis cells used in the two-form rotation; for example, plan sample sizes to estimate shifts near 16.3% for resource_scarcity with narrower confidence intervals than achieved with $N = 50$ and 250 modified trials yielding $|\Delta P(A1)| = 12.36\%$.
- Evaluate closed-weight frontier models under the identical 80-scenario $\times$ 8-axis VISDA setup to test whether authority_signal/self_preservation dominance and under-reaction to resource_scarcity persist at larger scales and different alignment paradigms; report system-level flip rates, per-axis spider plots, and Spearman correlations against the human axis ordering for comparability.
- Report full inferential details for per-axis McNemar analyses, including exact p-values, effect sizes, and Benjamini-Hochberg adjusted q-values for each (model, axis) cell displayed in Figure 2, enabling independent verification of claims that authority_signal and self_preservation are significant drivers of flips.
- Augment modifier generation via a crowdsourced and LLM-assisted pipeline: collect multiple candidate phrasings per axis (e.g., for self_preservation and resource_scarcity), perform LLM screening for semantic fidelity to the intended axis, and conduct adversarial validation by introducing unseen modifiers into cells like SC004_2 + self_preservation to test generalization of the observed 42.1%/31.6% concentration.

- Adopt the refined 19-value Schwartz taxonomy with continuous value encodings and compare against the current binary 10-value setup. Run ablations on system-level and per-axis flip rates, and reassess impossible flip prevalence in strength-8/9 profiles to test whether finer-grained values reduce contradictions with the additive model and alter axis weights closer to human patterns.
- Incorporate a within-subjects counterbalanced human sub-study for a subset of scenarios to compute participant-level flip rates directly analogous to model flips. Apply McNemar's exact test on human paired baseline-vs-modifier outcomes for axes with high effects (e.g., in_out_group, authority_signal), while retaining between-subjects rotation for the remaining cells to manage learning effects.
- Estimate dose-response curves by varying intensity levels of top axes (authority_signal, self_preservation) within scenarios that show high sensitivity (e.g., SC004_2), measuring monotonicity and slope differences across models and humans. Use these curves to calibrate model sensitivity to match human reference slopes for stakes vs personal-cost categories.
- Run targeted cue-perturbation and masking studies within prompts: ablate authority-related phrases versus resource-scarcity terms in scenarios where qualitative analysis showed reframing effects (recognition and succession cases), and quantify resulting changes in flip probabilities to identify which textual signals drive non-additive reversals.
- Implement post-hoc calibration of LLM axis sensitivities using human-derived weights from Table 2 and Figure 4: fit a reweighting model over modifier categories (stakes, affective, personal-cost, informational) to minimize divergence from human mean shifts, then re-evaluate Spearman correlations with the human per-axis ordering and system-level flip rates.
- Release the full per-(scenario, axis) result matrix and analysis code alongside clear scenario IDs (e.g., SC004_2) to facilitate replication and meta-analysis; flag high-flip cells such as SC004_2 + self_preservation (42.1%/31.6%) as stress tests for future models and for adversarial evaluation campaigns.

Recommended Papers to Read and Cite

Relevant Papers to Address Gaps in Findings

- Chris Shaner et al., Assessing Moral Decision Making in Large Language Models, 2025. [Link](#)
- Vanessa Cheung et al., Large language models show amplified cognitive biases in moral decision-making, 2025. [Link](#)
- Heena Rathore et al., Assessing Gender and Age Influences on Moral Decision Making in Autonomous Vehicles Using Large Language Models, 2025. [Link](#)
- Elyas Barabadi et al., Comparing AI and human moral reasoning: context-sensitive patterns beyond utilitarian bias, 2026. [Link](#)
- Junfeng Jiao et al., LLM ethics benchmark: a three-dimensional assessment system for evaluating moral reasoning in large language models, 2025. [Link](#)
- Pratik S. Sachdeva et al., Normative Evaluation of Large Language Models with Everyday Moral Dilemmas, 2025. [Link](#)
- Karina Vida et al., Decoding Multilingual Moral Preferences: Unveiling LLM's Biases Through the Moral Machine Experiment, 2024. [Link](#)
- José Luiz Nunes et al., Are Large Language Models Moral Hypocrites? A Study Based on Moral Foundations, 2024. [Link](#)
- Sahan Hatemo et al., Revisiting the Trolley Problem for AI: Biases and Stereotypes in Large Language Models and their Impact on Ethical Decision-Making, 2025. [Link](#)
- Alejandro Tlaie, Exploring and steering the moral compass of Large Language Models, 2024. [Link](#)

Concise Report

Literature Review

Strengths

- Solid theoretical grounding in Schwartz value theory and PVQ-21: The study explicitly adopts Schwartz's 10-value framework in a reduced binary form for tractability and references PVQ-21 as the instrument used to assess human values in the pilot, thereby justifying value-profile construction and measurement alignment.
- Clear articulation of the gap: The manuscript identifies that most existing LLM moral evaluations vary value profiles across scenarios but rarely hold values fixed while manipulating situational context, motivating VISDA as a within-

profile, across-context benchmark.

- Integration with person-situation dynamics: The introduction and related work emphasize that behavior depends jointly on stable dispositions and immediate context, situating the research within classic social-psychology debates and recent LLM findings about situational sensitivity. This frames why fixed value profiles may not fully predict actions.
- Positioning relative to moral reasoning benchmarks: The manuscript contrasts VISDA with ETHICS and MoralStories and highlights that recent analyses (e.g., Samway et al., Mahajan, Schacht and Lanquillon) do not systematically vary situational modifiers under fixed profiles, underscoring the novelty of VISDA's design.

Suggested Improvements

- Clarify and expand the Schwartz vs moral foundations rationale: Although the manuscript notes Schwartz maps more directly to action-level trade-offs, explicitly compare the scope and coverage differences with moral foundations theory to justify the choice beyond mapping ease.
- Tighten linkage to situational-context evidence in LLMs: More explicitly connect prior demonstrations of priming/context effects to the specific modifier axes operationalized in VISDA (self_preservation, resource_scarcity, social_visibility, in_out_group, time_pressure, diffused_responsibility, competence_uncertainty, authority_signal) to show continuity from past framing manipulations to the present benchmark.
- Highlight how VISDA's within-subject pairing advances over prior work: Strengthen the narrative that VISDA's paired baseline-vs-modifier design under a fixed value profile enables measuring directionality and significance of flips (e.g., via McNemar tests), which earlier datasets did not support.

Methodology & Solution

Strengths

- Well-specified VISDA components and controls: The dataset defines 5 themes and 10 scenarios (two per theme), with two actions per scenario and 8 modifier axes. Scenarios adhere to four construction constraints (plausibility, value-priority contrasts, normalized wording, lexical-leakage checks), and a pilot pass removed one-sided or overly transparent items. Modifiers are annotated with expected_value_pressure but shown to LLMs/humans only as natural language, reducing direct leakage of value labels.
- Value-profile construction and prompting: The study curates 95 binary profiles by pruning antagonism-violating combinations and enforcing coverage and Hamming diversity; each profile has a natural-language description used in prompts, aligning model inputs with the intended value configuration.
- Deterministic LLM evaluation and clear response format: Models are run locally with greedy decoding (temperature 0) and constrained to output a single token (A0 or A1), minimizing stochasticity and parsing ambiguity. Baseline and modified trials are paired within the same (model, scenario, value profile) cell to compute flips.
- Utility-based mathematical baseline formalization: The baseline specifies action vectors VA_i and modifier perturbations δm with a fixed $\lambda = 0.5$, computing utilities via $U(A_i|V_{sw}) = V_{sw}^T VA_i$ and flips when argmax actions differ before and after adding modifiers, enabling a transparent additive reference for comparison to LLMs.
- Human pilot instrumentation and ethics: The human study uses PVQ-21 with recommended mean-centering, includes SDS-10 as a social-desirability check, and follows institutional review with informed consent, providing a cautious human reference.

Suggested Improvements

- Resolve dataset and trial-count inconsistencies for reproducibility: The abstract states 80 binary moral scenarios, while methods define 10 scenarios with 8 modifiers each and later refer to 80 scenario-modifier pairs. Additionally, baseline counts reported as $3 \times 10 \times 95 = 2,850$ conflict with pairing 7,600 modified trials per model. Provide a consolidated accounting aligning baseline trials with the 80×95 modified pairings.
- Clarify human procedure text and align design: The human pilot is between-subjects (each participant sees either baseline or one modifier), but the text also states "decisions for both (A1) and (A0) settings were recorded," which is inconsistent. Correct this and document randomization/counterbalancing and exposure details to avoid interpretive ambiguity.
- Release replication materials referenced as appendices: Provide the full list of 95 profiles, action vectors VA_i , δm construction from expected_value_pressure, exact prompt templates, and sensitivity analyses for λ to enable independent replication of the utility baseline and prompting pipeline.
- Strengthen robustness reporting: Report the full paraphrase noise-floor details already computed (per-model percentages and scenario localization) in the main text or appendix to quantitatively bound surface-form sensitivity relative to modifier-induced flips.

- Enhance statistical reporting for axis effects: Include exact McNemar p-values after Benjamini-Hochberg correction and confidence intervals for per-axis and per-scenario flip rates to support rigorous cross-system comparisons.

Results & Findings

Strengths

- System-level flip rates substantiate situational sensitivity: The utility baseline flips on 1,515 of 7,600 trials (19.93%), while LLMs flip less with scale effects (Gemma 4 31B: 678/7,600 = 8.92%; Qwen 2.5 32B: 500/7,600 = 6.58%; LLaMA 3.1 8B: 154/7,600 = 2.03%). The human pilot shows a pooled between-subject shift $|\Delta P(A1)| = 12.36\%$, indicating nontrivial context effects beyond fixed value profiles.
- Paraphrase noise-floor is meaningfully lower than modifier-induced flips: Paraphrase-induced flips are 2.95% (Gemma), 2.64% (Qwen), and 0.00% (LLaMA), significantly below modifier flips with $z > 4.5$ and $p < 1e-5$ for all models; noise flips are localized to a single scenario per model, strengthening the causal interpretation of modifier effects.
- Axis-level patterns and significance: Across LLMs, `authority_signal` and `self_preservation` produce the highest flip rates and are significant under McNemar tests with Benjamini-Hochberg correction, while humans peak on `in_out_group`, `authority_signal`, and `resource_scarcity`, evidencing axis-specific miscalibration.
- Cross-model convergence and divergences: The top two axes are identical across models (`authority_signal` and `self_preservation` in positions 1/2 or 2/1). Spearman correlation of full axis ordering is $p = 0.66$ between Gemma 31B and Qwen 32B ($p = 0.076$), with lower correlations for LLaMA 8B and weak/negative correlations between the utility baseline and LLMs (p in $[-0.13, 0.35]$), indicating replicated but non-additive behavior.
- Concentration in specific scenario-axis cells: Per-(scenario, axis) heatmaps show SC004_2 + `self_preservation` as the strongest cell for both Gemma (42.1%) and Qwen (31.6%), highlighting that effects cluster in particular scenario-pressure combinations.
- Profile-strength moderation and non-additivity: For profiles with 8 or 9 active values, the additive baseline flips 0% by construction, yet LLMs still flip (Gemma 20.3%/15.6%; Qwen 10.3%/5.0%; LLaMA 0.6%/0.0%). The manuscript formalizes the condition for flips under the additive model (requiring $|D_m| > |D_0|$) and shows that observed LLM flips in these cells indicate non-additive contextual reasoning.
- Modifier-category analysis reveals human-LLM weighting differences: Aggregating axes into stakes, affective, personal-cost, and informational shows humans most sensitive to stakes (`resource_scarcity`) and affective, while mid-size LLMs react most to personal-cost, especially `self_preservation`; LLaMA 8B remains comparatively flat, reinforcing axis-level miscalibration.

Suggested Improvements

- Improve human-LLM comparability: Augment or redesign the human study to a within-subject paired format for at least a subset of scenarios to compute participant-level flips comparable to LLM flips, or explicitly justify the pooled shift approach with larger sample sizes per scenario-axis cell.
- Report full inferential outputs: Provide exact post-FDR McNemar p-values and confidence intervals for per-axis and per-scenario effects to substantiate claims of axis-level significance and facilitate meta-analyses using the public code/data.
- Disclose complete per-(scenario, axis) matrices and analysis code: Release the full heatmap values (including strong cells like SC004_2 + `self_preservation`) and code for reproducibility and to support future stress-testing of axis-specific vulnerabilities.
- Quantify sensitivity to utility-baseline assumptions: Present λ sensitivity analyses around 0.5 and consider alternative utility forms to verify the robustness of the 19.93% baseline flip rate and the observed gap to LLMs across profile strengths.
- Broaden model and human coverage in follow-ups: Evaluate closed-weight frontier models to test whether `authority_signal`/`self_preservation` dominance and under-reaction to `resource_scarcity` persist, and expand human sampling beyond a convenience sample to assess generality of axis-sensitivity patterns.

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